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RESEARCH ARTICLE

Optimizing Variable Admittance Control for Remote Ultrasound Scanning Under Uncertain Environment

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ABSTRACT Precise force control in remote robotic ultrasound systems is critical for optimizing image quality and ensuring patient safety. However, conventional admittance control strategies face limitations in achieving high-precision force tracking during interaction while maintaining accurate position tracking in free motion. To address this challenge, we propose an adaptive variable admittance approach integrated with a novel coarse-to-fine force control strategy, which requires online estimation of environmental properties. The environmental information is estimated by a fusion algorithm that combines force and position data from sensors with confidence scores derived from ultrasound images. Furthermore, a compensation term is introduced to the variable stiffness control law to mitigate estimation uncertainties, thereby enhancing force tracking accuracy. Additionally, an energy tank mechanism is implemented to guarantee system passivity under varying damping and stiffness conditions. The effectiveness of the proposed method was experimentally validated using a teleoperated ultrasound system, tested on both a vascular phantom model and human upper limb. The proposed controller demonstrated stable force tracking performance while maintaining compliance throughout the interaction process. The force tracking errors were maintained within ranges of 0.2 N and 0.4 N with standard deviation of 0.02 N and 0.2 N for the phantom and human experiments, respectively.

INDEX TERMS Variable admittance control, robotic ultrasound scanning, environment estimation, force tracking.

I. INTRODUCTION

Robotic Ultrasound systems (RUSS) have been extensively investigated over the past decade [1], [2], aiming to enhance scanning accuracy, motion repeatability and operational maneuverability in ultrasound image acquisition [3], [4]. The fundamental goal of robotic ultrasound scanning is to achieve precise force and pose control of the ultrasound transducer, which is mounted on the end-effector of a robotic manipulator teleoperated by a haptic device, as illustrated in Fig. 1. Due to the inherited characteristic of ultrasound imaging, image quality significantly depends on the acoustic coupling between the transducer and the human body, which is primarily determined by the normal force applied to the patient's surface [5], [6]. To achieve optimal image quality, the recommended force levels vary for different tissue types (2.0 N for thyroid examinations and 8.4 N for deeper tissues in abdomen) [7]. Maintaining a constant force is a crucial aspect of RUSS, as it prevents excessive pressure on the human body, thereby ensuring both safety and comfort during the scanning

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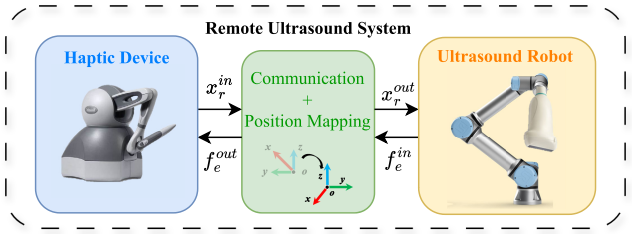


FIGURE 1. Schematic representation of bilateral control in a robotic ultrasound system, illustrating the bidirectional flow of position and force information.

procedure [8], [9], [10]. In manual ultrasound scanning, this is usually achieved through the experience of the human practitioner. Therefore, a dedicated force controller with compliance is essential for RUSS to produce consistent, non-deformed ultrasound images during interaction while safeguarding patient safety [11].

Hybrid force-position control approach [12] combines both force and position control law in two decoupled subspaces, which is particularly effective for tasks that require decoupled dynamic responses, such as grinding and polishing in industrial applications. For example, Gilbertson and Anthony [7] developed a 1 degree-of-freedom (DOF) hand-held robotic ultrasound probe using a hybrid position/force controller to maintain a constant probe-patient contact force. The controller was implemented by simplifying the contact model as two interfaces (human-machine and probe-patient) using a set of masses, springs, and dampers. Jiang et al. [13] implemented a compliant hybrid controller that integrates force control for maintaining orthogonal force with a position control for executing desired scan trajectory, to achieve automatic orthogonal probe positioning relative to the tissue surface. Normal direction detection was achieved using a model-based optimization method, enabling probe orientation computation independent of visual features. Additionally, Goel et al. [14] introduced a Bayesian Optimization framework for probe guidance on unknown surfaces, and implemented a hybrid pose and force controller to maintain consistent acoustic coupling. The force error, defined as the difference between expected and actual forces, served as input to a proportional controller that regulated the velocity of the end-effector. A key limitation of applying above force control law in the normal direction is the loss of position control functionality, which is essential for position mapping in free motion space of remote ultrasound systems. Moreover, transitioning between position and force control mode is not seamless and necessitates controller reparameterization, potentially leading to discontinuities in system behavior.

Admittance control have been extensively investigated in the field of physical human-robot interaction (pHRI) to achieve compliant force control and stable interaction behavior [15], [16]. The capability to regulate both motion and force in a unified framework makes these control approaches particularly suitable for our ultrasound scanning applications [17], [18]. Li et al. [19] proposed an adaptive variable admittance control strategy to ensure stability and compliance during interaction with both human and environment. Sufficient contact force is achieved through an adaptive term that adjusts the damping of the virtual damper based on the force exerted by environment. However, while the system achieves sufficient contact force, it cannot guarantee precise force value due to the variability introduced by human-applied force. For dynamic force tracking in uncertain environments, an adaptive variable impedance control method was researched by Duan et al. [20], which actively tracked the dynamic force and compensated for uncertainties. The force tracking performance exhibited significant overshoot during contact transitions and rapid changes in environment stiffness. Li et al. [21] employed fixed virtual stiffness and predefined target force values in the impedance control law to achieve compliant interaction with the lower back. However, quantifying the appropriate virtual stiffness values to achieve desired forces remained challenging. Wang et al. [22] extended conventional admittance control to a six-dimensional force regulation framework towards automated robotic ultrasound imaging. These approaches [20], [22], [23] aimed to achieve force tracking by eliminating the virtual stiffness term associated with position error from the admittance law, thus avoiding the need for accurate environment modeling. However, the absence of stiffness term compromises position tracking capability, which is crucial for integrating compliant interaction behaviors in teleoperation applications.

To the best of our knowledge, existing literature on force control strategies for RUSS rarely addresses stable force tracking for remotely position-controlled robots and often requires reparameterization due to unknown environmental information. Our work aims to design a unified compliant framework that supports both position and force tracking in respective scenarios. Therefore, this paper presents a novel adaptive variable admittance strategy for robotic ultrasound systems that achieves both pose tracking in free space and force tracking in contact space, while maintaining compliance and stability. As such, this article brings the following contributions:

- A unified compliant framework with adaptive variable admittance control for RUSS in both teleoperation and autonomous scenarios.
- A novel coarse-to-fine force control strategy that achieves desired contact force through online estimation of environmental location and stiffness.
- A fusion algorithm that integrates force and position data from sensors with confidence scores derived from ultrasound images to estimate environmental information.

The remainder of this paper will be organized as follows. The proposed adaptive variable admittance control strategy will be presented in Section II. Subsequently, the estimation method of environmental location and stiffness will be

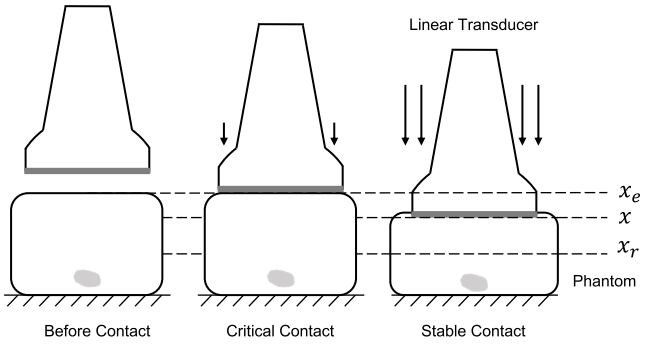


FIGURE 2. The interaction process between the transducer and the phantom involves three phases: before contact (*i.e.*, free motion), critical contact, and stable contact.The location of the phantom at the initial contact point is denoted as x_e, while x and x_r refer to the current and reference positions of the transducer, respectively.All positions are represented in one dimension for simplicity.

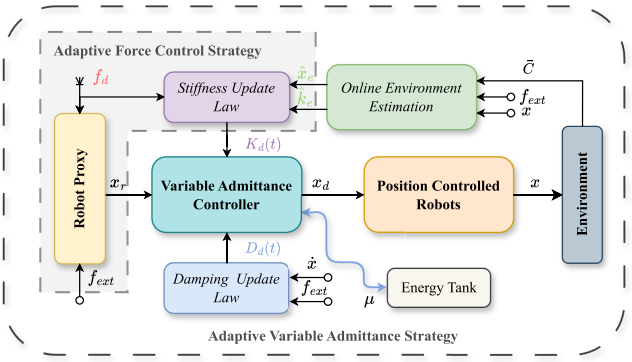


FIGURE 3. Block diagram of the proposed adaptive variable admittance strategy: robot proxy is responsible for receiving position commands from either remote site or local intelligence and publishing reference position x_r.

discussed in Section III. Finally, the experiments and results will be presented in Section IV followed by conclusion and discussion in Section V.

II. ADAPTIVE VARIABLE ADMITTANCE STRATEGY

A. PROBLEM STATEMENT

To achieve compliant force control during ultrasound scanning, impedance or admittance control is frequently utilized to regulate the interaction between the robot and the environment. The term admittance will be consistently used in this work owing to the position-controlled nature of our employed robot. To ensure a comprehensive understanding, we will briefly explain the admittance control law here. Consider a manipulator in contact with its environment at the end-effector, a prevalent form of admittance law in Cartesian space is given by

$$\boldsymbol{M}_d\ddot{\tilde{\boldsymbol{x}}} + \boldsymbol{D}_d\dot{\tilde{\boldsymbol{x}}} + \boldsymbol{K}_d\tilde{\boldsymbol{x}} = \boldsymbol{F}_{ext} \tag{1}$$

where $\tilde{\boldsymbol{x}} := \boldsymbol{x} - \boldsymbol{x}_r$ represents the pose error, defined as the difference between the current pose $\boldsymbol{x}$ and the reference pose $\boldsymbol{x}_r$. The pose of the end-effector $\boldsymbol{x}$ can be calculated using the forward kinematics map $f(\cdot)$ from the joint positions $\boldsymbol{q} \in \mathbb{R}^n$, *i.e.*, $\boldsymbol{x} = f(\boldsymbol{q})$. Desired virtual inertia, damping, and stiffness matrices are denoted by $\boldsymbol{M}_d, \boldsymbol{D}_d, \boldsymbol{K}_d \in \mathbb{R}^{6\times6}$ in operational space, respectively. The external force $\boldsymbol{F}_{ext}$ is measured by a 6 DOF force/torque sensor mounted between the end-effector of the robot and the ultrasound transducer.

The interaction between the transducer and the human body during ultrasound scanning can be characterized by three distinct phases, as depicted in Fig. 2. To facilitate the subsequent discussion, we will focus on the normal direction of the interaction as the normal force is the primary component that significantly influences the imaging quality of ultrasound scanning. All subsequent notations are represented in one dimension, without loss of generality. During free motion, the admittance controller can asymptotically track the reference trajectory regardless of the selected stiffness value k_d, thus providing an effective alternative for compliant position tracking in remote ultrasound scanning tasks.

To explain the interaction behavior in contact space mathematically, we simplify the model of environment as a linear spring with stiffness k_e. The contact force is readily represented as $f_e = k_e(x_e - x)$. Rewriting the contact force in terms of the pose error $\tilde{x}$ yields

$$f_e = k_e(x_e - x) = k_e x_e - k_e(\tilde{x} + x_r) \tag{2}$$

Substituting $g(s) = \frac{1}{ms^2+ds+k}$ derived from the Laplace transform into above equation yields

$$(ms^2 + ds + k + k_e)f_e = (ms^2 + ds + k)k_e(x_e - x_r) \tag{3}$$

The steady-state contact force is then obtained as

$$f_{e,ss} = \frac{k_e k}{k_e + k}(x_e - x_r) \tag{4}$$

Given known environmental parameters k_e and x_e, it is straightforward to calculate the reference position x_r, with a fixed virtual stiffness k to achieve the desired force f_d, as

$$x_r = x_e - \frac{k_e + k}{k_e k} f_d \tag{5}$$

Similarly, the desired force f_d can also be achieved by tuning the virtual stiffness k with a fixed reference position x_r, as

$$k = \frac{k_e f_d}{(x_e - x_r)k_e - f_d} \tag{6}$$

From the above analysis, precise force control requires appropriate selection of virtual stiffness k and reference position x_r, which is challenging due to their coupled relationship. Furthermore, environmental parameters are typically unknown and difficult to measure accurately in advance. The assumption of a linear environment model may not be sufficiently reliable, particularly for soft tissues [24].

B. VARIABLE DAMPING STRATEGY

Conventional admittance control often struggles to adapt to the varying requirements of different scenarios when applying constant parameters. This limitation becomes particularly evident during transitions from free space to contact space [19]. To address this issue, a variable damping strategy is introduced to dynamically adjust the damping effect based on the specific requirements of different phases during interaction. The admittance model can be rewritten by

$$\boldsymbol{M}_d(t)\ddot{\tilde{\boldsymbol{x}}} + \boldsymbol{D}_d(t)\dot{\tilde{\boldsymbol{x}}} + \boldsymbol{K}_d(t)\tilde{\boldsymbol{x}} = \boldsymbol{F}_{ext} \tag{7}$$

where $\boldsymbol{M}_d(t)$, $\boldsymbol{D}_d(t)$ and $\boldsymbol{K}_d(t)$ are time-varying inertia, damping and stiffness matrices, respectively. We assume that they are symmetric and positive-definite for all $t \geq 0$, in order to preserve their physical meaning. Constant desired inertia is used, *i.e.*, $\boldsymbol{M}_d(t) = \boldsymbol{M}_c$, to ensure the best tradeoff between stability and compliance [25].

The proposed variable damping law is designed as

$$\boldsymbol{D}_d(t) = diag(\boldsymbol{\mathcal{D}}_c - \alpha|\dot{\boldsymbol{x}}| + \eta|\boldsymbol{F}_{ext}|) \tag{8}$$

where $\boldsymbol{\mathcal{D}}_c \in \mathbb{R}^6$ represents the initial value of damping in the vector form. The notation $diag(\cdot)$ denotes the diagonal matrix operation that constructs a diagonal matrix from a vector. The positive constants α and η control the damping variation *w.r.t.* the velocity $\dot{\boldsymbol{x}}$ and external contact force $\boldsymbol{F}_{ext}$.

In the design of the damping law, the first velocity-related term has minimal impact during interaction with environment, as velocity remains generally low. During free motion, the damping effect diminishes due to higher velocity, enabling large movements and ensuring rapid responsiveness in position tracking. The second force-related term which adjusts the damping based on the external force $\boldsymbol{F}_{ext}$, becomes active only in contact space, as external force remains zero before the robot makes contact with the environment. Hence, before the contact is established, the damping effect is primarily governed by the velocity-related term. Once contact is established, it becomes dominant in stabilizing the controller, since a relatively large coefficient will be chosen for the consideration of safety. In this manner, the proposed variable damping law ensures low damping during free motion for rapid pose tracking, and high damping during contact to ensure compliant interaction behavior. Besides, a smooth transition from non-contact to contact is also achieved without compromising system stability. The passivity-preserving property of the variable damping term still guarantees the stability of the control system, which will be discussed later in this section.

C. FORCE CONTROL STRATEGY

Inspired by the manual operations of sonographers, who continuously adjust the force exerted on the probe to find optimal imaging conditions, an intuitive approach for our admittance controller is to adjust the reference position to identify an appropriate interaction plane. However, the exact value of x_r calculated using (5) often fails to reach the desired contact force due to inappropriate virtual stiffness and inaccurate environment parameters. Based on these observations, we regard the adjustment of reference position as a coarse-tuning tool, primarily aimed at locating an approximate contact plane for sufficient interaction. Meanwhile, the adjustment of virtual stiffness functions as a fine-tuning step to achieve a more precise force control.

Algorithm 1 Adaptive Force Control Strategy

```
// Initialization
x_r = x_{r,p}, k_d = k_{init};
Loop:
    online estimation to obtain x̂_e and k̂_e;
    if f_e < λ f_d then
        // Coarse searching step
        x_{r,t} = x_{r,t-1} + v_r · δt ;
        k_{d,t} = k̂_{e,t-1}
    else
        // Fine tuning step
        k_{d,t} = k̂_e f_d / ((x̂_e − x_{r,t}) k̂_e − f_d)
    end if
end loop
```

Building upon the above insights, we propose an adaptive force control strategy that integrates coarse searching and fine-tuning steps to achieve stable force tracking as delineated in Alg. 1. The strategy incorporates online estimation of environmental parameters without relying on precise model accuracy. The fundamental principle underlying the coarse step is to iteratively search for a coarse reference position that approximates the desired force f_d, bounded by an upper limit λf_d. The environment parameters are exactly estimated during this searching process, and the virtual stiffness k is assigned the value of the estimated stiffness $\hat{k}_e$. This assignment ensures the stiffness matching between the controller and the environment, thereby mitigating the risk of potential unstable contact. Once the measured force reaches the predefined threshold, indicating that the robot has identified a supporting contact plane, the algorithm transitions to the fine-tuning phase. In this phase, the applied force is precisely regulated by a modulated virtual stiffness. Given the acquired environmental information and reference position, the desired stiffness can be computed as follows:

$$k_d = \frac{\hat{k}_e f_d}{(\hat{x}_e - x_r)\hat{k}_e - f_d} \tag{9}$$

Note that a stiffness smoother is consistently employed to mitigate sudden fluctuations in virtual stiffness throughout the entire coarse-to-fine process. Theoretically, the desired stiffness calculated by (9) should direct the robot towards achieving the desired force. However, due to inherent uncertainty in environmental parameter estimation and the limitations of the linear model assumption, the actual force may deviate from the desired value. To address this

discrepancy, a compensation term is introduced to augment the desired stiffness, thereby enhancing force tracking performance. This compensation term is proportional to the difference between the desired force and external force with a proportional coefficient β, expressed as

$$k_{d,t} = k_{d,t-1} + \beta(f_d - f_e) \tag{10}$$

The desired stiffness calculated by (9) establishes a baseline stiffness value for achieving the target force. The subsequent feedback term ensures that force tracking error remains within a prescribed range of the target value. The compensation term is bounded to prevent excessive stiffness values that could compromise the robot's compliance.

D. ENERGY TANK BASED PASSIVITY

A significant limitation arising from the introduction of variable stiffness is the loss of passivity [26]. Consider the total energy of the system as a non-negative storage function:

$$\mathcal{V}(\tilde{\boldsymbol{x}}, \dot{\tilde{\boldsymbol{x}}}) = \frac{1}{2}\dot{\tilde{\boldsymbol{x}}}^T \boldsymbol{M}_c \dot{\tilde{\boldsymbol{x}}} + \frac{1}{2}\tilde{\boldsymbol{x}}^T \boldsymbol{K}_d(t)\tilde{\boldsymbol{x}} \tag{11}$$

With a slight abuse of notation, we will hereafter use $\mathcal{V}(t)$ to indicate the value of $\mathcal{V}(\tilde{\boldsymbol{x}}, \dot{\tilde{\boldsymbol{x}}})$ at time t for simplicity.
Considering our time-invariant inertia $\boldsymbol{M}_c$ and time-varying stiffness $\boldsymbol{K}_d(t)$, the time derivative of $\mathcal{V}(t)$ is given by

$$\dot{\mathcal{V}}(t) = \dot{\tilde{\boldsymbol{x}}}^T \boldsymbol{M}_c \ddot{\tilde{\boldsymbol{x}}} + \tilde{\boldsymbol{x}}^T \boldsymbol{K}_d(t)\dot{\tilde{\boldsymbol{x}}} + \frac{1}{2}\tilde{\boldsymbol{x}}^T \dot{\boldsymbol{K}}_d(t)\tilde{\boldsymbol{x}} \tag{12}$$

Therefore, replacing admittance law (7) in the above yields

$$\dot{\mathcal{V}}(t) = \dot{\tilde{\boldsymbol{x}}}^T \boldsymbol{F}_{ext} + (-\dot{\tilde{\boldsymbol{x}}}^T \boldsymbol{D}_d(t)\dot{\tilde{\boldsymbol{x}}} + \frac{1}{2}\tilde{\boldsymbol{x}}^T \dot{\boldsymbol{K}}_d(t)\tilde{\boldsymbol{x}}) \tag{13}$$

Indeed, the system would be passive *w.r.t.* the input/output pair $(\boldsymbol{F}_{ext}, \dot{\tilde{\boldsymbol{x}}})$ only if the term between the brackets was negative, ensuring energy dissipation within the system. Specifically, the energy dissipated by the damping term should surpass the energy injected by the stiffness term, in order to satisfy the passivity condition. However, due to the loss of passivity, stable interaction in contact space and asymptotic pose tracking in free space are no longer guaranteed.

To restore the passivity of the system, an energy tank is incorporated into the admittance model as shown in Fig. 3. This tank stores dissipated energy from the model and subsequently utilizes it to implement behaviors that might otherwise threaten passivity. The tank dynamics is given by

$$\dot{x}_t = \frac{\mu}{x_t}\tilde{\boldsymbol{p}}^T \boldsymbol{M}_c^{-1} \boldsymbol{D}_d(t) \boldsymbol{M}_c^{-1}\tilde{\boldsymbol{p}} - \frac{\kappa^T}{x_t}\dot{\tilde{\boldsymbol{x}}} \tag{14}$$

where $x_t \in \mathbb{R}$ and $T(x_t) = \frac{1}{2}x_t^2$ represent the tank state and tank energy function, respectively. The parameter $\mu \in \{0, 1\}$ functions as a switch, which disables the storage of dissipated energy generated by the damping term when tank energy reaches the maximum capacity T_{max}.

The stiffness-related term κ is defined as

$$\kappa(t) = \begin{cases} -\boldsymbol{K}_v(t)\tilde{\boldsymbol{x}} & \text{if } T(x_t) > \varepsilon \\ 0 & \text{otherwise} \end{cases} \tag{15}$$

where $\boldsymbol{K}_v(t) = \boldsymbol{K}_d(t) - \boldsymbol{K}_c$ represents the time-varying component extracted from the stiffness matrix. The parameter ε is a small positive value that defines the minimum energy level required to activate the energy tank. The underlying mechanism for detaching stiffness term like this stems from treating the variable stiffness term as an internal interaction between the robot and the energy tank, which maintains passivity when considering the entire system.

Proof of Passivity: Consider the energy storage function $\mathcal{Q}(\tilde{\boldsymbol{x}}, \tilde{\boldsymbol{p}}, x_t)$ for the entire system as

$$\begin{aligned} \mathcal{Q}(t) &= H(\tilde{\boldsymbol{x}}, \tilde{\boldsymbol{p}}) + T(x_t) \\ &= \frac{1}{2}\tilde{\boldsymbol{p}}^T \boldsymbol{M}_c^{-1}\tilde{\boldsymbol{p}} + \frac{1}{2}\tilde{\boldsymbol{x}}^T \boldsymbol{K}_c\tilde{\boldsymbol{x}} + \frac{1}{2}x_t^2 \end{aligned} \tag{16}$$

where $\tilde{\boldsymbol{p}} = \boldsymbol{M}_c\dot{\tilde{\boldsymbol{x}}}$ is the momentum of the robot. The time derivative of the energy storage function $\mathcal{Q}(t)$ is given by:

$$\begin{aligned} \dot{\mathcal{Q}}(t) &= \left(\boldsymbol{F}_{ext}^T - \dot{\tilde{\boldsymbol{x}}}^T \boldsymbol{D}_d(t)\right)\dot{\tilde{\boldsymbol{x}}} + \mu\tilde{\boldsymbol{p}}^T \boldsymbol{M}_c^{-1} \boldsymbol{D}_d(t) \boldsymbol{M}_c^{-1}\tilde{\boldsymbol{p}} \\ &= \boldsymbol{F}_{ext}^T\dot{\tilde{\boldsymbol{x}}} - \tilde{\boldsymbol{p}}^T \boldsymbol{M}_c^{-1}(\boldsymbol{D}_d(t) - \mu\boldsymbol{D}_d(t))\boldsymbol{M}_c^{-1}\tilde{\boldsymbol{p}} \end{aligned} \tag{17}$$

Since $\mu \in \{0, 1\}$ and $\boldsymbol{D}_d(t) > 0$, then we have $(1-\mu)\boldsymbol{D}_d(t) \geq 0$ implying that

$$\mathcal{Q}(t) - \mathcal{Q}(0) \leq \int_0^t \boldsymbol{F}_{ext}^T(\tau)\dot{\tilde{\boldsymbol{x}}}(\tau)d\tau \tag{18}$$

The passivity conditions are satisfied through the incorporation of the energy tank, enabling the implementation of variable admittance interaction model while maintaining the passivity constraint [27].

III. ONLINE ENVIRONMENT ESTIMATION

In this section, an online estimation method is introduced to estimate the contact location x_e and stiffness k_e of the environment. We fuse confidence data derived from ultrasound image and force readings from sensor to detect initial contact location and employ a Kalman filter to estimate environmental stiffness from real-time force and position data recorded by the robot.

A. ESTIMATION OF ENVIRONMENT LOCATION

The location information serves as an indicator of whether the robot is in contact with the environment. The contact point can be detected using only force sensor readings, with a force threshold. However, manually setting a force threshold may not adequately adapt to changes in different environments and lacks robustness in determining the contact state. Ultrasound images provide 2D real-time visualization of tissue information, which may be a more effective option for contact detection. Therefore, we propose a contact detection algorithm that fuses force sensor readings with ultrasound confidence data. This approach allows for a more accurate and robust determination of the contact point without relying on predefined force threshold.

The quality of ultrasound images is measured by confidence map, which aims at estimating the per-pixel quality of

Algorithm 2 Fusion Algorithm for Contact Detection

Input : $\bar{c}, f_z$ in stream
Output: contact point: p_{ct}
// calculate standard score online
Function `isOutlier(`x`)`**:**
  update μ and σ with x using Welford's method;
  $z_{score} \leftarrow (x-\mu)/\sigma$ **if** $z_{score} > 3$ **then**
    return *true*;
  else
    return *false*;
  end if
End Function
while *true* **do**
  $c_{sm}, f_{sm} \leftarrow$ `KalmanFilter(`$\bar{c}, f_z$`)`;
  flag_contact $\leftarrow$ `isOutlier(`c_{sm}`)` && `isOutlier(`f_{sm}`)`;
  if *flag_contact* **then**
    $p_{ct} \leftarrow f(q)$; // record position
    break;
  end if
end while

the ultrasound signal. Here, the confidence map is computed by integrating the pixel intensity along the scan line [28] as

$$C(u,v) = 1 - \left(\sum_{v=0}^{AN-1} I^2(u,v)\right)^{-1} \sum_{v=0}^{v} I^2(u,v) \tag{19}$$

where $C(u,v)$ denotes the confidence value at pixel (u,v) and $I(u,v)$ represents the intensity of the ultrasound image at the same pixel location. Additionally, AN denotes the number of pixels along the scan line, *i.e.*, the height of the image. The confidence value is normalized and falls within the range of $[0,1]$.

The mean confidence value is then calculated by averaging the confidence values of all pixels in the confidence map:

$$\bar{C}(t) = \frac{1}{AN \cdot LN} \sum_{u=0}^{LN-1} \sum_{v=0}^{AN-1} C(u,v) \tag{20}$$

where $\bar{C}(t)$ represents the mean confidence value and LN denotes the number of scan lines, *i.e.*, the width of image.

Both force and confidence values are filtered by a Kalman filter to reduce the noise and then the contact point is estimated by Alg. 2. Instead of selecting specific thresholds for force and confidence values, a common statistical probability of 3σ is used to unify them. The contact point is then estimated as the point where both force readings and the confidence values exceed this 3σ threshold. To achieve this, an online normalization method [29] is applied to the force and confidence values to detect outliers.

B. ESTIMATION OF ENVIRONMENT STIFFNESS

The estimation of environment stiffness is inherently imprecise due to the simplification of the environment model.

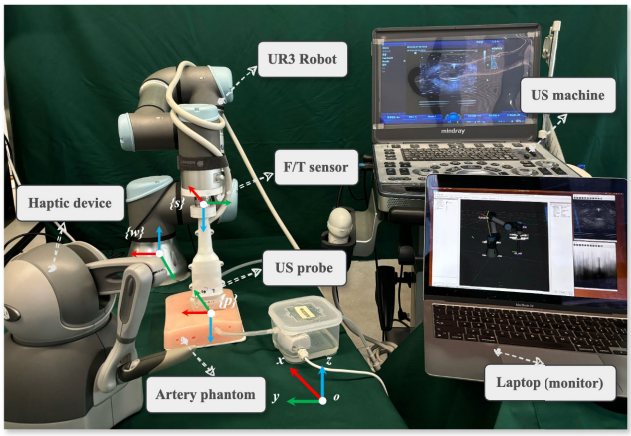


FIGURE 4. An illustration of the ultrasound robotic system.

However, this imprecision becomes acceptable for controller performance when appropriate compensation methods are employed. Consequently, we apply a second-order Kalman filter to estimate the environment stiffness from real-time force and position data recorded by the robot.

Considering contact only in z direction, let $\boldsymbol{f}_n \doteq [f_z, \dot{f}_z]^T$ be the state vector, the state transition matrix $\mathcal{8}$ is defined as $\mathcal{8} = \left[\begin{smallmatrix} 1 & \Delta z \\ 0 & 1 \end{smallmatrix}\right]$, where Δz is the position interval between two consecutive samples. Then we have

$$\begin{pmatrix} f_{z,n+1} \\ \dot{f}_{z,n+1} \end{pmatrix} = \begin{pmatrix} 1 & \Delta z \\ 0 & 1 \end{pmatrix} \begin{pmatrix} f_{z,n} \\ \dot{f}_{z,n} \end{pmatrix} \tag{21}$$

For fast convergence, the initial value of $\dot{f}_{z,0}$ is set to 0.3 N/mm and the Kalman filter operates at a frequency of 25 Hz to meet real-time requirement. As detailed derivation of Kalman filter is beyond the scope of this paper, equations for covariance extrapolation, update, and Kalman gain calculation are omitted here.

IV. RESULTS

A. EXPERIMENTAL SETUP

As shown in Fig. 4, the proposed remote ultrasound robotic system comprises a 6-DOF robotic manipulator (UR3e, Universal Robots), a 6-axis force/torque sensor (Nano25, SI-125-3, ATI), an ultrasound machine (M8, Mindray) with a linear transducer (L12-4s, Mindray), a commercial artery phantom (LydMed, CN) and a 6-DOF haptic device (Touch, 3D Systems) for teleoperation. The UR3 robot is controlled through the Robot Operating System (ROS2), running on a computer equipped with an Intel i9-10900X CPU, and an NVIDIA GeForce RTX 3070 GPU, with Ubuntu 20.04 system. The force/torque sensor is connected to the computer via Ethernet and operates at a sampling rate of 1 kHz. The B-mode ultrasound images are captured from the ultrasound machine using a frame grabber (CSZX, CN) at a frequency of 25 Hz, with a resolution of 550 × 520. The maximum allowable contact force is limited to 12 N.

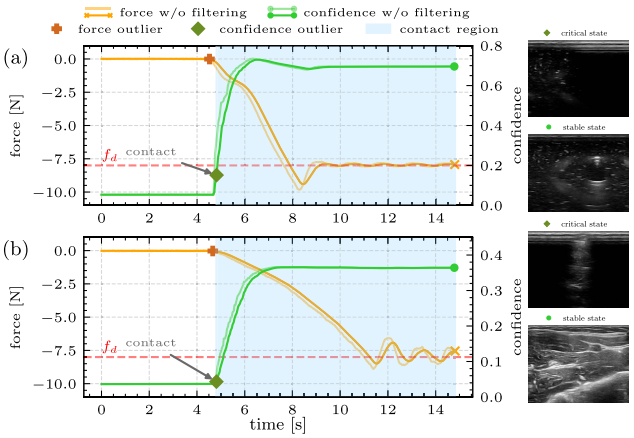


FIGURE 5. The estimation process performed on (a) phantom and (b) human upper limb. The fusion approach identifies the confidence outlier point as the contact point and subsequently estimates the stiffness value using force and position data within the contact region.Two representative frames showing critical and stable states are illustrated in the right column for both cases.

TABLE 1. Environment Estimation Results Conducted on Phantom and Human Upper Limb with 5 Samples.

Type	Mean ± Standard Deviation (Range)	
	Location [mm]	Stiffness [N/mm]
Phantom	**195.28 ± 0.0748** (195.17 ∼ 195.38)	**1.9697 ± 0.0127** (1.9495 ∼ 1.9697)
Human	**211.00 ± 0.2403** (210.73 ∼ 211.40)	**1.1315 ± 0.0306** (1.0892 ∼ 1.1773)

B. ENVIRONMENT ESTIMATION RESULTS

To demonstrate the effectiveness of the proposed fusion algorithm, environment estimation experiments were conducted on both a phantom and a human upper limb with the desired contact force set to 8 N. As shown in Fig. 5, the force and confidence data initially remained static before the robot made contact with the environment. Subsequently, a force outlier was detected at around 4.5 s, followed by a confidence outlier shortly after. The fusion algorithm identified the confidence outlier point as the contact point in both phantom and human cases. The corresponding contact region was highlighted by a blue area extending from this point. The algorithm achieves robust contact detection by fusing two complementary contact-perceiving data, even when the force data are contaminated by external disturbances.

Once the contact point was detected, the current position of the end-effector was recorded as x_e and another stiffness estimation thread was activated to estimate the stiffness online. The estimation results were collected from 5 samples of experiments for each case, and the average values were reported in Table 1. The location results for the phantom showed an average value of 195.28 mm, with a standard deviation of 0.0748 mm. For the human upper limb, the average value was 211.00 mm, with a standard deviation of 0.2403 mm. The error range, within 0.21 mm and 0.67 mm for these two cases, is acceptable considering the vessel diameter of approximately 3 mm. The lower accuracy in location estimation for the human compared to the phantom is expected, since the contact and motion characteristics of the human limb are more complex. The average stiffness values for the phantom and human upper limb were 1.9697 N/mm and 1.1315 N/mm, with a standard deviation of 0.0127 N/mm and 0.0306 N/mm, respectively. The stiffness estimation results for the human limb exhibited greater variation compared to those for the phantom due to the natural movements, such as vasodilation and muscle contraction. These inherent estimation errors can be mitigated by the compensation term in the proposed strategy.

C. FORCE CONTROL PERFORMANCE

To demonstrate the force control performance, the robot manipulator was first teleoperated to a specific position above the target tissue. Subsequently, the local-autonomy mode, which employs the proposed adaptive force control strategy, was activated to initiate interaction with the target tissue. For the following experiments, the virtual mass and initial virtual damping ratio were set to 6.25 kg and 0.7. The coefficients α and η for variable damping term were set to 250.0 and 10.0, respectively. The desired force f_d was set to 8.0 N and the force threshold λ was set to 0.9. The compensation gain β for stiffness variation was set to 50.0. The initial virtual stiffness was configured at 0.5 N/mm, a value significantly lower than the estimated stiffness, to minimize the impact of the initial stiffness selection.

The entire interaction processes for both phantom and human upper limb were recorded and presented in Fig. 6. Prior to reaching the left boundary of the blue area, the robot tracked the reference trajectory without any external force. Once the left boundary was reached, *i.e.*, the contact was detected, the robot entered a transition phase from non-contact to contact. At this juncture, the variable damping strategy functioned to stabilize the interaction behavior, with the damping increasing significantly in response to the external force. Simultaneously, the robot's velocity was reduced to adapt to the new interaction dynamics. The blue zone indicates the duration when the controller implemented a coarse searching strategy, specifically, adjusting the reference trajectory to identify an approximate contact location. The virtual stiffness was set to match the estimated stiffness value to ensure stable interaction behaviors. Excessively high virtual stiffness reduces the compliance of the robot, whereas too small value may introduce unexpected oscillations.

After exiting the blue area, the robot entered the yellow zone, where the fine-tuning strategy was implemented to adjust the virtual stiffness. As indicated within the red dashed rectangle region in Fig. 6, the virtual stiffness was modulated to achieve the target contact force. The variation in stiffness values exhibits subtle differences between phantom and human cases. In the phantom case, the desired stiffness, calculated from (9) is lower than the current virtual stiffness. Consequently, the virtual stiffness was decreased to align with

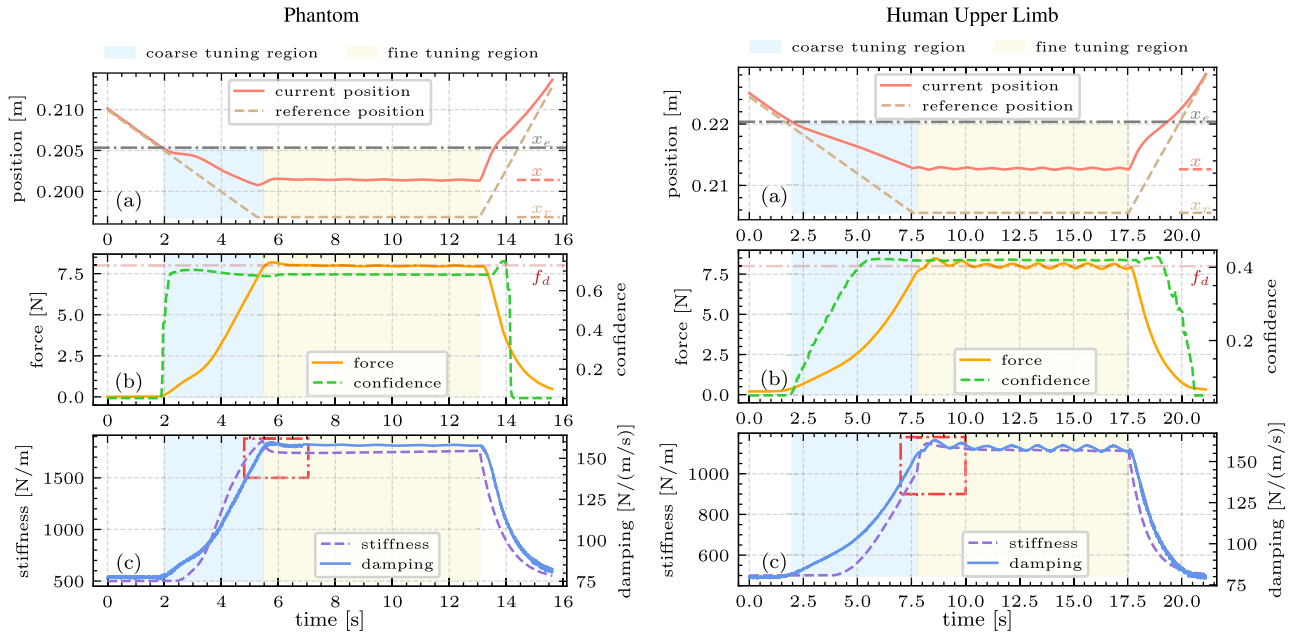


FIGURE 6. The interaction behaviors of the proposed controller when interacting with the (left) phantom and (right) human upper limb.The blue area employs the coarse searching approach to identify a rough contact location, while the yellow area applies the fine-tuning strategy which modulates the admittance stiffness to achieve desired force f_d.The red dashed rectangle highlights the region where the specific stiffness variation mechanism operates.

the calculated desired stiffness. Conversely, in the human case, the desired stiffness exceeds the current virtual stiffness, resulting in a sharp increase of the virtual stiffness towards the desired value. These contrasting scenarios demonstrate the capability of controller to adapt to diverse contact conditions, particularly varying stiffness levels.

Throughout the remainder of fine tuning region, the compensation term functions to mitigate force tracking error arising from inaccurate environment estimation and external disturbances. In the phantom case, the desired stiffness appears to be accurate to achieve the target force, resulting in minimal impact of the compensation term on force control. However, in the human case, the desired stiffness induced a relatively large force deviation from the desired force. This deviation is likely attributable to inaccuracies in environment estimation for complicated human tissue structures. Consequently, the compensation term functioned to reduce the virtual stiffness and thereby mitigate the force tracking error. The observed force fluctuations in the human case resulted from vasodilation and muscle contraction of the limb tissue.

A detailed and quantitative analysis of the force tracking error is presented in Fig. 7. To analyze the force tracking error, the reference desired force signal was fitted using a first-order polynomial function and processed by a low-pass filter. Three target force levels (6, 8, 10) N were evaluated on both the phantom and human upper limb. The force error behavior in the blue area (coarse region) was not analyzed, as no standardized desired force existed during this transition period. The force error distribution in the yellow

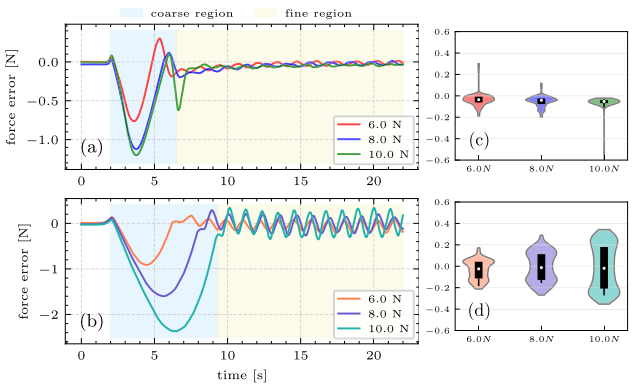


FIGURE 7. The force tracking error of the proposed controller during interaction with the (a) phantom and (b) human upper limb. The violin plots (c) and (d) illustrate the error distribution in the fine tuning region for each case.

area (fine tuning region) was illustrated in the violin plots as shown in Fig. 7 (c) and (d). In the phantom case, the force tracking error consistently remained within range of 0.2 N across all target force levels. In the human case, the force error also maintained a near-zero mean value and remained within range of 0.4 N. The mean and standard deviation of force tracking error measured during stable interaction periods were reported in Table 2. In the phantom case, the force tracking error remained relatively stable, with the mean value staying within 0.05 N and the standard deviation within 0.02 N. In contrast, the human case exhibited similar mean value but significant fluctuations, with the standard

TABLE 2. The Mean and Standard Deviation of Force Tracking Error Measured During Stable Interaction Periods, Specifically from 12 to 20 seconds in Fig. 7.

Figure 7	Mean ± Standard Deviation	
Class	Phantom [N]	Human [N]
6 N	-0.0234 ± 0.0206	-0.0553 ± 0.0874
8 N	-0.0319 ± 0.0191	-0.0313 ± 0.1367
10 N	-0.0495 ± 0.0130	-0.0133 ± 0.2050

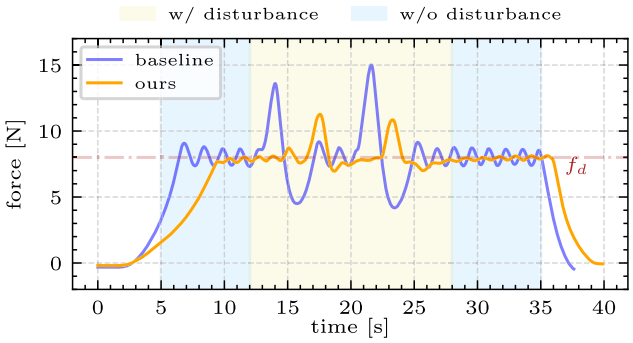


FIGURE 8. The force control behaviors of the proposed controller compared with baseline method, when unpredicted human movement occurs. Both controllers were tested on the human upper limb with a target force of 8 N.

deviation increasing proportionally to the target force level. This behavior can be attributed to enhanced vasodilation and muscle contraction in the human limb under increased contact forces.

To demonstrate the anti-disturbance performance of the proposed controller when faced with unpredictable human movement, we conducted a comparative experiment on the human upper limb against the existing method [14], as shown in Fig. 8. The proposed controller demonstrates smaller variations than the baseline method in the blue area (without disturbance), indicating superior force control performance. In the yellow area (with disturbance), the baseline method exhibits significant oscillations when encountering similar human movement. In contrast, the proposed controller responds rapidly to disturbances and recovers to the desired force level with minimal overshoot. This enhanced performance can be attributed to the damping effects of the admittance controller.

D. PASSIVITY BEHAVIOR

The energy tank remains active throughout free motion and contact phases. During the free motion phase, the energy flow primarily consists of storing energy dissipated by the damping term, while the stiffness term maintains a constant value. Potentially passivity-violating behavior occurs during the contact phase, where variations in the stiffness term are initiated. Here, the capacity of the energy tank T_{max} was set to 2.0 J with a minimum energy level ε of 0.01 J. The initial state of the tank $T(0)$ was set to 0.005 J. The energy tank behavior during interaction with the phantom and human upper limb is

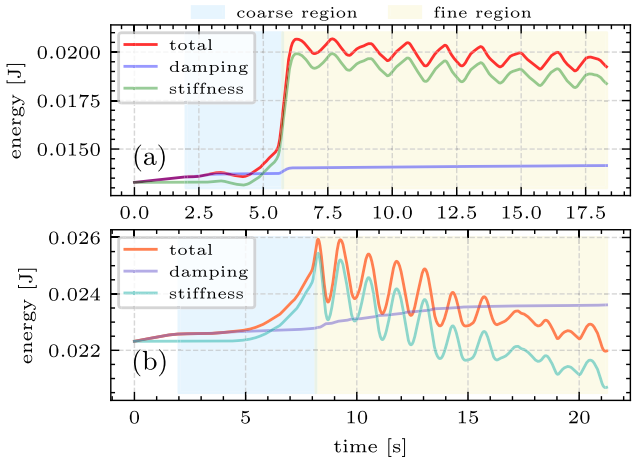


FIGURE 9. The behavior of energy tank while interacting with the (a) phantom and (b) human upper limb.The energy flows attributable by damping and stiffness terms are theoretically separated for visualization purpose.

presented in Fig. 9. The total energy stored in the tank remains within prescribed bounds, while the energy curves associated with damping and stiffness effects are shown separately to visualize their individual contributions. The energy curve associated with the damping term exhibits relatively low and smooth variations, corresponding to the low velocities maintained during interaction. The energy curve associated with the stiffness term, however, demonstrates more pronounced variations, particularly when transitioning from the coarse to fine region. These variations are primarily due to the joint effects of adaptive stiffness and admittance model state $\tilde{\boldsymbol{x}}$ and $\dot{\tilde{\boldsymbol{x}}}$.

V. CONCLUSION AND DISCUSSION

This work presents a novel adaptive variable admittance strategy for remote robotic ultrasound systems, achieving desired force tracking without compromising pose tracking capability. The proposed strategy integrates a variable damping law with a coarse-to-fine force control strategy to ensure stable force tracking and safe interaction, while adapting to the unknown soft tissue properties. Benefiting from the incorporation of energy tank, the passivity of the whole system is guaranteed. The experimental results demonstrate that the proposed controller maintains stable force tracking over extended periods, significantly reducing hand fatigue for operators. The force tracking errors observed in the experiments (maximum error $0.055 \pm 0.2N$) represent a substantial improvement over freehand ultrasound scanning procedures, which typically show deviations of $0.76 \pm 0.63N$ [7]. This level of precision is clinically important, as the force control capability of the proposed controller ensures patient safety by maintaining contact force within comfortable limits. Such precision is essential for force-sensitive applications like elastography imaging, where consistent tissue compression is essential for accurate tissue characterization and high-quality diagnostic images

[30], [31]. We believe that the controller design can be readily integrated into existing robotic ultrasound systems to enhance normal force tracking performance. Besides, this general framework can be extended to multi-dimensional force control. The current investigation focuses on implementing different force level control mechanisms while optimizing image quality based on specific force levels. Further work will be conducted to determine the optimal force level for different target tissues based on the ultrasound image quality metrics. Additionally, clinical validation will be conducted on a specific clinical scenario under medical supervision to assess real-world applicability.

REFERENCES

[1] Y. Fu, W. Lin, X. Yu, J. J. Rodríguez-Andina, and H. Gao, "Robot-assisted teleoperation ultrasound system based on fusion of augmented reality and predictive force," *IEEE Trans. Ind. Electron.*, vol. 70, no. 7, pp. 7449–7456, Jul. 2023.

[2] F. Suligoj, C. M. Heunis, J. Sikorski, and S. Misra, "Robust–An autonomous robotic ultrasound system for medical imaging," *IEEE Access*, vol. 9, pp. 67456–67465, 2021.

[3] Z. Jiang, S. E. Salcudean, and N. Navab, "Robotic ultrasound imaging: State-of-the-art and future perspectives," *Med. Image Anal.*, vol. 89, Oct. 2023, Art. no. 102878.

[4] K. Li, Y. Xu, and M. Q.-H. Meng, "An overview of systems and techniques for autonomous robotic ultrasound acquisitions," *IEEE Trans. Med. Robot. Bionics*, vol. 3, no. 2, pp. 510–524, May 2021.

[5] D. Raina, S. Chandrashekhara, R. Voyles, J. Wachs, and S. K. Saha, "Robotic sonographer: Autonomous robotic ultrasound using domain expertise in Bayesian optimization," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, May 2023, pp. 6909–6915.

[6] K. Su, J. Liu, X. Ren, Y. Huo, G. Du, W. Zhao, X. Wang, B. Liang, D. Li, and P. X. Liu, "A fully autonomous robotic ultrasound system for thyroid scanning," *Nature Commun.*, vol. 15, no. 1, p. 4004, May 2024.

[7] M. W. Gilbertson and B. W. Anthony, "Force and position control system for freehand ultrasound," *IEEE Trans. Robot.*, vol. 31, no. 4, pp. 835–849, Aug. 2015.

[8] M. Tirindelli, M. Victorova, J. Esteban, S. T. Kim, D. Navarro-Alarcon, Y. P. Zheng, and N. Navab, "Force-ultrasound fusion: Bringing spine robotic-U.S. to the next 'level,'" *IEEE Robot. Autom. Lett.*, vol. 5, no. 4, pp. 5661–5668, Oct. 2020.

[9] D. Huang, C. Yang, M. Zhou, A. Karlas, N. Navab, and Z. Jiang, "Robot-assisted deep venous thrombosis ultrasound examination using virtual fixture," *IEEE Trans. Autom. Sci. Eng.*, vol. 22, pp. 381–392, 2025.

[10] X. Yan, S. Luo, Y. Jiang, M. Yu, C. Chen, S. Zhu, G. Huang, S. Song, and X. Li, "A unified interaction control framework for safe robotic ultrasound scanning with Human-Intention-Aware compliance," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, Oct. 2024, pp. 14004–14011.

[11] R. Tsumura, T. Tomioka, Y. Koseki, and K. Yoshinaka, "Safe contact force generation for robotic thyroid ultrasound imaging," *IEEE Robot. Autom. Lett.*, vol. 9, no. 2, pp. 1700–1707, Feb. 2024.

[12] M. H. Raibert and J. J. Craig, "Hybrid Position/Force control of manipulators," *J. Dyn. Syst., Meas., Control*, vol. 103, no. 2, pp. 126–133, Jun. 1981.

[13] Z. Jiang, M. Grimm, M. Zhou, Y. Hu, J. Esteban, and N. Navab, "Automatic force-based probe positioning for precise robotic ultrasound acquisition," *IEEE Trans. Ind. Electron.*, vol. 68, no. 11, pp. 11200–11211, Nov. 2021.

[14] R. Goel, F. Abhimanyu, K. Patel, J. Galeotti, and H. Choset, "Autonomous ultrasound scanning using Bayesian optimization and hybrid force control," in *Proc. Int. Conf. Robot. Autom. (ICRA)*, May 2022, pp. 8396–8402.

[15] F. Ficuciello, L. Villani, and B. Siciliano, "Variable impedance control of redundant manipulators for intuitive human–robot physical interaction," *IEEE Trans. Robot.*, vol. 31, no. 4, pp. 850–863, Aug. 2015.

[16] C. Liu, G. Peng, K. Zhao, J. Li, and C. Yang, "Neural learning-based adaptive force tracking control for robots with finite-time prescribed performance under varying environments," *IEEE Trans. Ind. Electron.*, vol. 71, no. 12, pp. 16338–16347, Dec. 2024.

[17] A. Q. Keemink, H. V. D. Kooij, and A. H. A. Stienen, "Admittance control for physical human–robot interaction," *Int. J. Robot. Res.*, vol. 37, no. 11, pp. 1421–1444, Apr. 2018.

[18] F. Ferraguti, C. T. Landi, L. Sabattini, M. Bonfè, C. Fantuzzi, and C. Secchi, "A variable admittance control strategy for stable physical human–robot interaction," *Int. J. Robot. Res.*, vol. 38, no. 6, pp. 747–765, Apr. 2019.

[19] H.-Y. Li, I. Paranawithana, L. Yang, T. S. K. Lim, S. Foong, F. C. Ng, and U.-X. Tan, "Stable and compliant motion of physical human–robot interaction coupled with a moving environment using variable admittance and adaptive control," *IEEE Robot. Autom. Lett.*, vol. 3, no. 3, pp. 2493–2500, Jul. 2018.

[20] J. Duan, Y. Gan, M. Chen, and X. Dai, "Adaptive variable impedance control for dynamic contact force tracking in uncertain environment," *Robot. Auto. Syst.*, vol. 102, pp. 54–65, Apr. 2018.

[21] K. Li, J. Wang, Y. Xu, H. Qin, D. Liu, L. Liu, and M. Q.-H. Meng, "Autonomous navigation of an ultrasound probe towards standard scan planes with deep reinforcement learning," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, May 2021, pp. 8302–8308.

[22] J. Wang, C. Lu, Y. Lv, S. Yang, M. Zhang, and Y. Shen, "Task space compliant control and six-dimensional force regulation toward automated robotic ultrasound imaging," *IEEE Trans. Autom. Sci. Eng.*, vol. 21, no. 3, pp. 3652–3663, Jul. 2024.

[23] Z. Huo, H. Wu, R. Xu, M. Yuan, and X. Zhang, "Toward eFAST autonomous robotic ultrasound imaging: System integrations and experimental studies," *Sci. China Inf. Sci.*, vol. 68, no. 3, Feb. 2025, Art. no. 139203.

[24] J.-L. Gennisson, T. Deffieux, M. Fink, and M. Tanter, "Ultrasound elastography: Principles and techniques," *Diagnostic Interventional Imag.*, vol. 94, no. 5, pp. 487–495, May 2013.

[25] S. Xiao, C. Wang, Y. Shi, J. Yu, L. Xiong, C. Peng, and L. Yang, "Visual optimization of ultrasound-guided robot-assisted procedures using variable impedance control," in *Proc. WRC Symp. Adv. Robot. Autom. (WRC SARA)*, Sep. 2021, pp. 128–133.

[26] F. Ferraguti, C. Secchi, and C. Fantuzzi, "A tank-based approach to impedance control with variable stiffness," in *Proc. IEEE Int. Conf. Robot. Autom.*, May 2013, pp. 4948–4953.

[27] F. Ferraguti, N. Preda, A. Manurung, M. Bonfè, O. Lambercy, R. Gassert, R. Muradore, P. Fiorini, and C. Secchi, "An energy tank-based interactive control architecture for autonomous and teleoperated robotic surgery," *IEEE Trans. Robot.*, vol. 31, no. 5, pp. 1073–1088, Oct. 2015.

[28] P. Chatelain, "Quality-driven control of a robotized ultrasound probe," Ph.D. dissertation, Fakultät für Informatik, Technische Universität München, Munich, Germany, 2016.

[29] B. P. Welford, "Note on a method for calculating corrected sums of squares and products," *Technometrics*, vol. 4, no. 3, p. 419, Aug. 1962.

[30] R. G. Barr and Z. Zhang, "Effects of precompression on elasticity imaging of the breast: Development of a clinically useful semiquantitative method of precompression assessment," *J. Ultrasound Med.*, vol. 31, no. 6, pp. 895–902, Jun. 2012.

[31] L. Sandrin, M. Tanter, J.-L. Gennisson, S. Catheline, and M. Fink, "Shear elasticity probe for soft tissues with 1-D transient elastography," *IEEE Trans. Ultrason., Ferroelectr., Freq. Control*, vol. 49, no. 4, pp. 436–446, Apr. 2002.



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